Md Obayed Reza Hridoy

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Working on embedded systems and firmware, embedded AI, and TinyML, with research in deep learning and computer vision for image-based detection on resource-constrained hardware.

Education

BRAC University — Dhaka, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering — *2025 – Present*

University of the People — Online
Associate Degree in Computer Science — *2026 – Present*

Experience

Co-Founder, Head of Research & Management — 2022 – Present
Tornetec — *Dhaka, Bangladesh*

- Leading research, strategic planning, and team management for an IT startup focused on AI integration into everyday human life.
- Overseeing prototype development and ensuring alignment with real-world technology needs.

Undergraduate Researcher — 2025 – 2026
Laboratory of Space System Engineering Technology (LaSSET), BRAC University — *Dhaka, Bangladesh*

- Working on computer vision and image-based detection tasks involving deep learning and CNN-based approaches.
- Developing and evaluating detection-based solutions for research applications using Python, MATLAB, Simulink, and PyTorch.

Founder & Manager — 2022 – 2024
Zyphrst

- Authorized NFT Selling Studio at Rarible Marketplace.
- Conducting in-depth market research to identify trends and opportunities in the NFT space.

Projects

GreenSync | *Co-Founder, Systems Architect & Lead Researcher* — 2025 – Present

- Leading project planning, research coordination, and development of a technology-driven smart waste management system utilizing a reward-based user engagement model.
- Serving as the founding Project Manager and Co-Founder while contributing to ongoing research activities.
- University Innovation Hub Program (UIHP) & Hult Prize – Grant-Awarded Project.

BRACU Alter | *Co-Lead, AI & Vision Sub Team* — 2025 – 2026

- Working on deep learning and computer vision for image-based detection in disaster environments for rover navigation.
- Training and fine-tuning models for real-time multi-class object detection on embedded vision hardware.

Achievements

Champion, Ideation 5.0
Second Runner-Up, Hult Prize – BRAC University 2025
National Round Participant, Hult Prize Bangladesh

Technical Skills

Languages: C/C++, Python, Java
Engineering Tools: MATLAB, Simulink, PSpice
AI & Computer Vision: Computer Vision, PyTorch, Deep Learning, CNN, Image/Object Detection
Embedded AI: ONNX Model Conversion, Model Deployment on Embedded Devices
Research: Data Analysis and Visualization, Quantitative Research Methods, Research Planning & Execution
Professional: Project Management, Interdisciplinary Collaboration, Strategic Planning